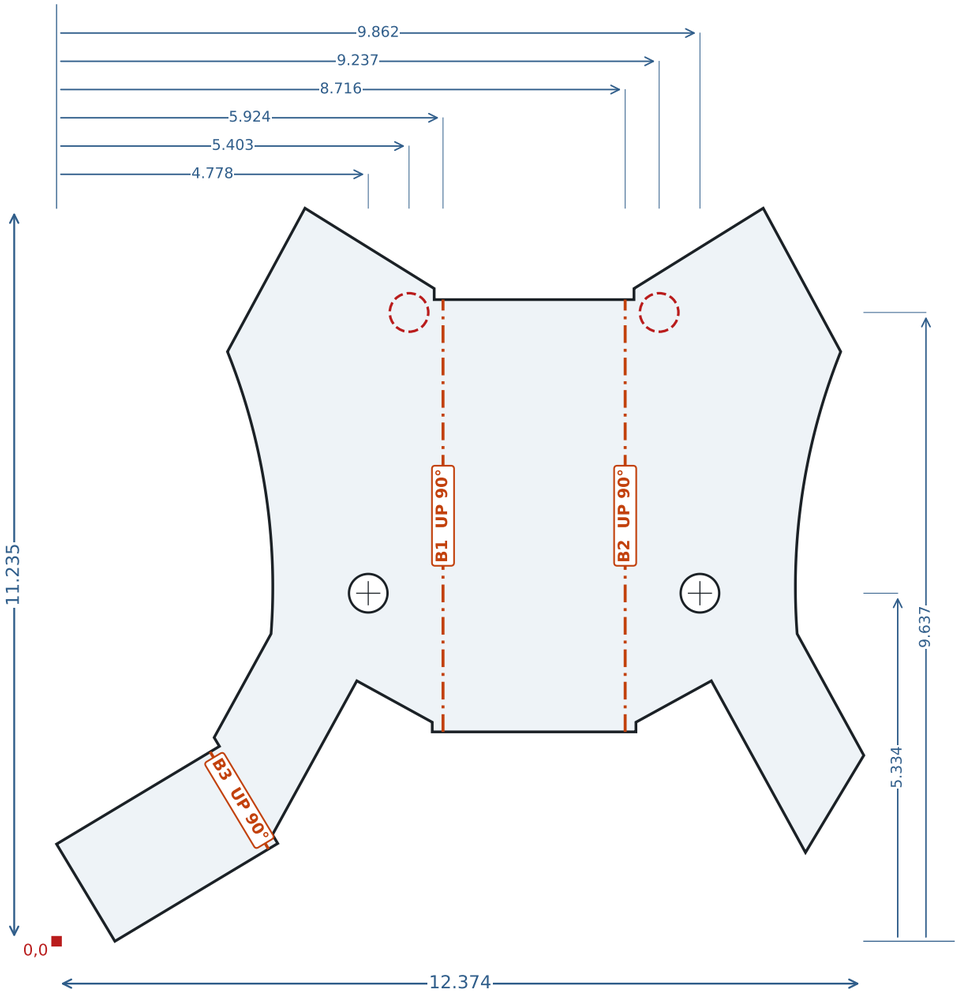
FR-10-1 · Part 1 "U-body" 2 sides + wall A-H + flange E-F (3 bends)

11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 12.374″ × 11.235″ · ≈ 2.29 lb each · QTY 1 FR + 1 FL = 2 per vehicle

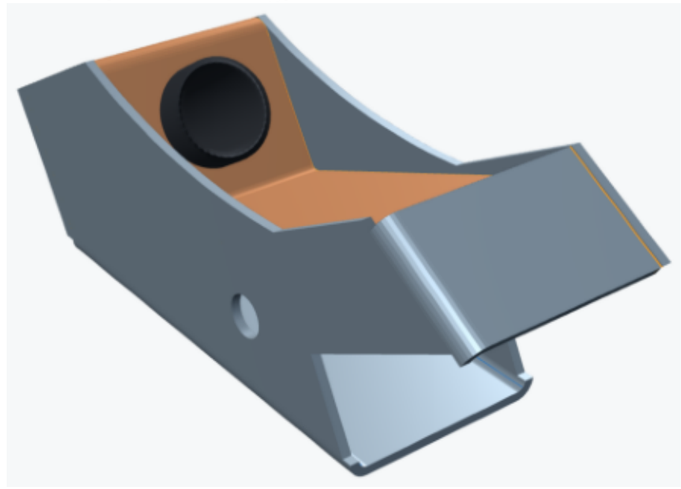
MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196″ (3.04 mm)

FLAT PATTERN (as laser-cut, marked side up)



FR-10-1 · 11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS

FORMED PART (after bending)



(shape; thickness now 11 ga 0.120″)

BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	UP (toward)	R=t 0.120	2	X = 5.924
B2	90.0°	UP (toward)	R=t 0.120	3	X = 8.716
B3	90.0°	UP (toward)	R=t 0.120	1	(2.354, 2.903) (3.249, 1.413)

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	Ø0.591	4.778	5.334	laser
H2	Ø0.591	9.862	5.334	laser
D1	Ø0.591	5.403	9.637	DRILL LATER
D2	Ø0.591	9.237	9.637	DRILL LATER

FILES

- FR-10-1_FR_cut.dxf (right, qty 1 / vehicle)
- FR-10-1_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-10-1_ref.dxf (reference: + bend lines, drill-later, material notes)

NOTES

1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. MATERIAL HR steel sheet A1011 CS, 11 ga t = 0.1196″ (3.04 mm). Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut - drill after bending/welding.
4. Deburr all edges, remove dross.
5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.

* bends 90° up, R = t; the 2 Ø0.591 holes near A (½" from A-B) are not in the flat - drill after bending (DRILL LATER)

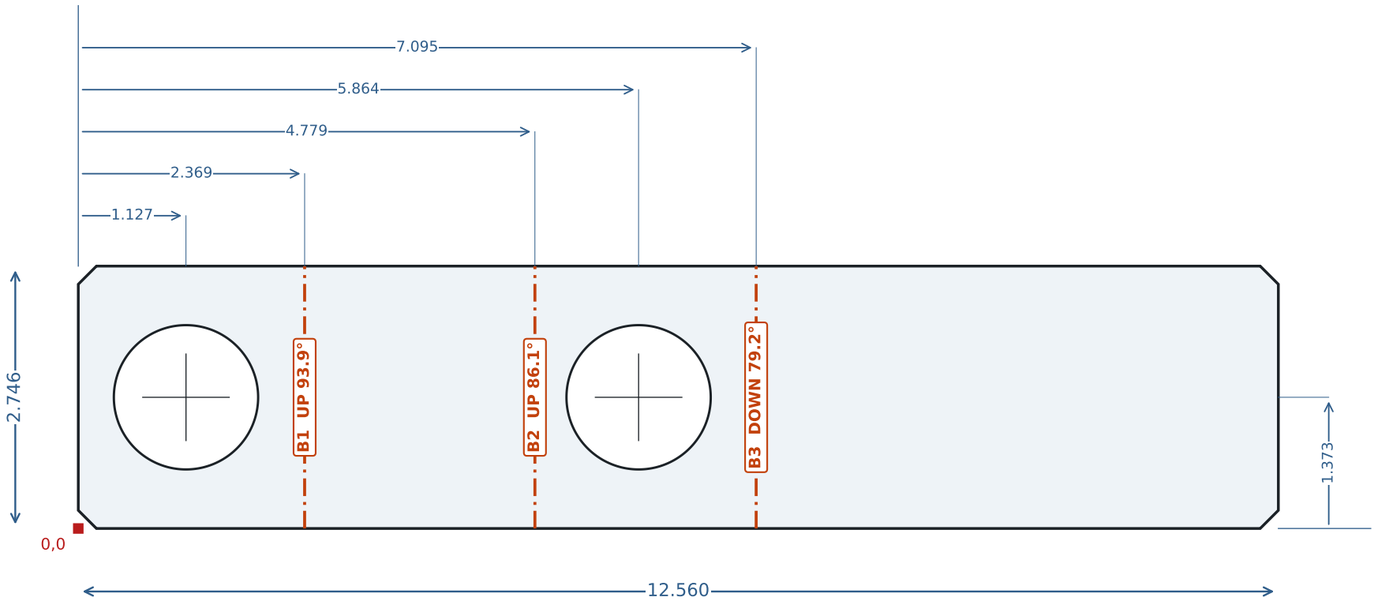
FR-10 Bracket: U-body + strip + tube	**HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)**	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV B**
FR-10-1 · Part 1 "U-body" 2 sides + wall A-H + flange E-F (3 bends)	**QTY 1 FR + 1 FL = 2 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-31	**SHEET 1/3**

FR-10-2 · Part 2 "strip" A-B · B-C · shelf · inner plate (3 bends)

11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 12.560″ × 2.746″ · ≈ 1.05 lb each · QTY 1 FR + 1 FL = 2 per vehicle

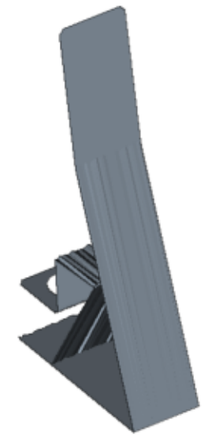
MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196″ (3.04 mm)

FLAT PATTERN (as laser-cut, marked side up)



FR-10-2 · 11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS

FORMED PART (after bending)



BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	93.9°	UP (toward)	R=t 0.120	5	X = 2.369
B2	86.1°	UP (toward)	R=t 0.120	6	X = 4.779
B3	79.2°	DOWN (away)	R=t 0.120	4	X = 7.095

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	Ø1.510	1.127	1.373	laser
H2	Ø1.510	5.864	1.373	laser

FILES

- FR-10-2_FR_cut.dxf (right, qty 1 / vehicle)
- FR-10-2_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-10-2_ref.dxf (reference: + bend lines, drill-later, material notes)

NOTES

1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. MATERIAL HR steel sheet A1011 CS, 11 ga t = 0.1196″ (3.04 mm). Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut - drill after bending/welding.
4. Deburr all edges, remove dross.
5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.

* X 79.2° opposite way; B 93.9°, C 86.1° up; corners 3/16"×45°; strip width = 3" - 2t - 0.015 (11 ga), from FR10_Bent model

FR-10 Bracket: U-body + strip + tube	**HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)**	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV B**
FR-10-2 · Part 2 "strip" A-B · B-C · shelf · inner plate (3 bends)	**QTY 1 FR + 1 FL = 2 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-31	**SHEET 2/3**

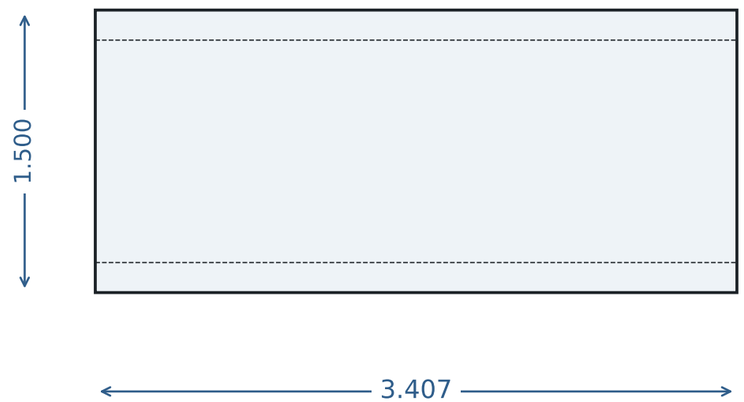
FR-10 · tube cut list

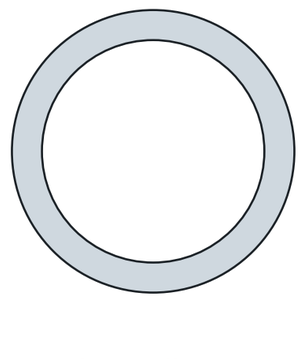
MATERIAL: DOM round steel tube A513

WALL THICKNESS: OD 1.500 × ID 1.181 (wall 0.160″)

Tube OD 1.5 / ID 1.181 cut 3.407″ (through A-B and the shelf)

Stock: DOM round steel tube OD 1.500 × ID 1.181 (wall 0.160″)





section, wall 0.1595″

Cut length 3.407″ ±1/32″, square ends, deburr

QTY 1 per side × 2 (FR + FL) = 2 per vehicle

FR-10 Bracket: U-body + strip + tube	**DOM round steel tube A513**	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV B**
Tube cut list	**QTY 1 per side × 2 (FR + FL) = 2 per vehicle**	Tol.: cut ±1/32″	2026-10-31	**SHEET 3/3**